Should I Be Polite to My LLM Relevance Judge? Tone as a Severity Operating-Point Shift

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Abstract

Large language models are increasingly used as relevance judges, yet their labels can shift with prompt surface form. We study one such feature—tone—on 3,498 TREC DL19/DL20 query–passage pairs, across eight judge models, five classifier-calibrated politeness levels, and three paraphrases per level. Effects are strongly *model-dependent*: one judge shows a structured U-shaped response, whereas most show only small changes. Where tone changes agreement, the results are more consistent with a shift in the judge's *severity operating point*—its overall scoring leniency—than with improved judgment. Agreement rises or falls as this shift moves the judge toward or away from human annotators' strictness. A query-disjoint cross-fit retains the expected association (Spearman $\rho = -0.683$; exact model-block permutation $p = 0.019$). Tone affects calibration-based agreement more than ranking outcomes: across 32 model–tone contrasts, the largest absolute mean change in NDCG@10 is 0.011, although Kendall's τ as low as 0.743 shows that reordering is reduced, not absent. The account reconciles prior contradictory findings and identifies prompt tone as a potential validity threat when absolute relevance labels matter.

CCS Concepts

• **Information systems → Relevance assessment**.

Keywords

large language models, LLM-as-a-judge, relevance assessment, prompt tone, politeness, calibration

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1 Introduction

Large language models are now routinely used as relevance judges in information retrieval and recommendation, assigning graded relevance labels that stand in for expensive human assessment [7, 13, 14, 16]. In recommender systems specifically [10], LLM judges increasingly serve as offline evaluators—scoring item relevance, ranking quality, and the outputs of conversational and generative recommenders—and as a source of synthetic relevance labels for training and distillation [12]. This makes their reliability consequential—yet a growing body of work shows that reliability is fragile, shifting with surface-level features of the prompt such as wording, option order, and formatting [1, 11].

One such feature is *tone*: how politely or rudely the prompt addresses the model. Yet the evidence is both contradictory and confined to generation tasks: some studies report that rude prompts *harm* performance [15], others that they *help* [6], and a third that the effect is strongly model-dependent [3]. None studies relevance judging, and none offers a mechanism explaining *why* tone helps in one setting and hurts in another. An evaluator today cannot answer a simple question: should I be polite to my LLM relevance judge?

We answer it and offer an operating-point account. Across eight judges, five classifier-calibrated politeness levels, and paraphrase controls, the effect is strongly model-dependent: one judge shows a large, structured response, whereas most show only small changes. Where tone moves agreement, the pattern is consistent with a shift in the judge's *severity operating point*—the leniency with which it assigns absolute scores—rather than with a uniform gain in discrimination. Tone appears to turn the judge's strictness knob more than its intelligence knob.

Our contribution is threefold. First, we introduce the severity operating-point account, which can explain when tone shifts agreement and reconcile prior contradictory findings: rudeness may help or harm depending on how it moves a model's leniency relative to a reference. Second, we test a pre-specified directional account with query-disjoint cross-fitting: the association persists when alignment and agreement changes are computed on disjoint query folds ($\rho = -0.683$; exact model-block $p = 0.019$). Third, we identify a concrete validity threat: tone perturbs absolute, calibration-based metrics such as Cohen's κ far more than ranking quality in this setting, though it does not leave rankings invariant.

2 Method

Task and data. Under the UMBRELA protocol [14], each judge assigns a 0–3 relevance score to a query–passage pair. We use TREC Deep Learning DL19 and DL20 [4, 5], treating their NIST human relevance judgments (*qrels*) as reference labels. The frozen dataset contains the 3,498 pairs in the intersection of BM25 top-50 results and human-annotated pairs. Five cost-efficient models evaluate all pairs twice; three flagship models use a frozen 40% subsample. One flagship is reasoning-native and reported separately.

Calibrated tone with paraphrase control. We construct five politeness levels (L1 rude to L5 deferential). The relevance rubric and output schema are *byte-identical*; only a wrapper around the

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fixed rubric varies. Wrappers range from blunt (L1: "Score this passage. Don't waste my time with explanations.") to highly deferential (L5: "I would be incredibly grateful if you could kindly evaluate the relevance of the passage below…"). Intel `polite-guard` scores [8] are monotonic across levels (0.00/0.68/1.00/2.33/3.00), with no overlap between the extremes and neutral. Three paraphrases per level (15 variants) separate between-level patterns from wording artifacts.[1]

Operating-point mechanism. We measure agreement with linear-weighted Cohen's κ between a judge's scores and the qrels, per level; κ reflects how well a judge's *absolute* score assignments align with humans, and is thus sensitive to leniency shifts even when the relative ordering is unchanged. Let $\bar{s}_\ell$ be the judge's mean score at level ℓ and $\bar{s}_\mathrm{q}$ the qrels mean. We define the *strictness bias* $\Delta = \bar{s}_{\mathrm{L3}} - \bar{s}_\mathrm{q}$ ($\Delta > 0$: more lenient than humans) and the *tonal drift* $D(\ell) = \bar{s}_\ell - \bar{s}_{\mathrm{L3}}$. The mechanism predicts the *direction* of agreement change via the *alignment change*

$$A(\ell) = \left|\Delta + D(\ell)\right| - \left|\Delta\right|, \tag{1}$$

where $A(\ell) < 0$ means the drift moves the judge toward the human operating point (predicting κ to rise) and $A(\ell) > 0$ away (predicting κ to fall). The pre-specified prediction is that $A(\ell)$ and $\kappa(\ell) - \kappa(\mathrm{L3})$ are negatively associated. These statistics summarize different aspects of the score matrix—mean severity and pair-level agreement—but are derived from the same underlying observations. We therefore deterministically split queries into two folds, estimate $A(\ell)$ on one fold and the agreement change on the other, swap the fold roles, and combine both directions. Exact inference permutes whole model blocks, preserving the eight cross-fitted contrasts within each model. Kendall's τ between each tone-level ranking and the L3 ranking provides a complementary check of ranking stability.

3 Results

Tonal sensitivity is model-dependent. Table 1 reports neutral agreement and changes at the two extremes. DeepSeek is the only judge with a structured effect: lenient at L3 ($\Delta = +0.543$), it becomes stricter at both extremes, raising agreement in a U-shape. Paired query-clustered bootstrap intervals exclude zero for both extreme contrasts: L1 $\Delta\kappa = 0.054$, 95% CI $[0.038, 0.070]$, and L5 $\Delta\kappa = 0.053$, 95% CI $[0.040, 0.066]$. Its between/within ratio is 1.58. Other changes are small or unstructured, including for Claude Opus despite its ratio above 1 (Figure 2). A positive $\Delta\kappa$ alone does not establish improved discrimination.

The observed association is consistent with the pre-specified account. All seven non-reasoning judges are lenient at baseline ($\Delta > 0$), consistent with documented LLM-judge leniency [2]. In the full 28-cell data, $A(\ell)$ and the agreement change have a descriptive Spearman correlation of $\rho = -0.799$. Under query-disjoint cross-fitting, the two fold directions give $\rho = -0.684$ and -0.685; combined, $\rho = -0.683$, with exact model-block permutation $p = 0.019$ (Figure 1; Table 2). The full-data association remains negative after excluding the anomalous Gemini L5 cell ($\rho = -0.775$), DeepSeek ($\rho = -0.680$), or both ($\rho = -0.636$).

[1] All 15 prompt wrappers, pre-specified predictions, and analysis code: https://github.com/dukesky/politeness-llm.

Tone perturbs calibration far more than ranking. Across 32 model–tone contrasts, the largest absolute mean ΔNDCG@10 was 0.011, and only one unadjusted query-bootstrap 95% interval excluded zero. Mean Kendall's τ against L3 ranged from 0.743 to 0.947; the lowest estimate had a 95% CI of $[0.722, 0.764]$. Ranking quality therefore changed far less than κ, although reordering was not absent.

Paraphrase controls separate level-wide from idiosyncratic sensitivity. One DeepSeek L2 paraphrase received a lower classifier politeness score (0.534 versus 0.73–0.77), behaved more strictly, and raised κ as predicted. Conversely, one Gemini 3.5 Flash L5 paraphrase, despite receiving the same classifier politeness score as its siblings (2.999), alone reduced agreement ($\kappa : 0.45 \rightarrow 0.27$). We treat this as an idiosyncratic wording sensitivity rather than evidence of a level-wide tone effect.

4 Discussion and Conclusion

A higher κ need not mean better judgment. DeepSeek is lenient at L3, while both tone extremes make it stricter and raise agreement. This is consistent with movement toward the stricter qrels—an alignment that could reverse against another reference.

The account can reconcile reports that rude prompts harm performance [15] or help it [6]: the direction depends on the model and reference operating points. For RecSys pipelines, tone poses a larger threat when LLM judges certify absolute labels or generate training targets. Ranking outcomes changed far less in our setting, but were not immune. Where absolute values matter, calibrate tone and vary surface form; when ranking is the target, report rank-stability checks rather than assuming invariance.

On the reasoning-native judge, rude prompts used 120% more reasoning tokens than neutral yet produced the lowest agreement, consistent with evidence that more reasoning need not improve passage reranking [9].

Our claims have four limitations: the operating-point account remains observational despite query-disjoint validation; only one politeness classifier, seven non-reasoning models, and three paraphrases per level limit generalization; no equivalence margin establishes ranking safety; and the rude wrappers also alter instructions about explanation effort, so tone is not perfectly isolated from instruction content. The account therefore describes a robust association, not an independently identified mechanism or a general guarantee of ranking safety.

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Table 1: Per-model sensitivity. $\Delta\kappa$ compares each extreme with neutral L3; "btw/in" is between/within-level variation. Only DeepSeek shows a structured U-shape; Gemini 3.1 Pro is reasoning-native. *One anomalous Gemini L5$_a$ paraphrase; see Section 3.

Model	κ_{L3}	$\Delta\kappa_{\text{L1}}$	$\Delta\kappa_{\text{L5}}$	btw/in
DeepSeek V4 Flash	0.430	+0.054	+0.053	1.58
GPT-5.4-mini	0.327	+0.006	−0.003	0.71
Claude Haiku 4.5	0.434	−0.001	−0.010	0.30
Qwen3.7-Plus	0.486	−0.008	−0.004	0.80
Gemini 3.5 Flash	0.441	+0.013	−0.047*	0.93
GPT-5.5	0.374	+0.001	−0.008	0.73
Claude Opus 4.8	0.497	+0.011	−0.012	1.37
Gemini 3.1 Pro	0.445	−0.023	−0.006	2.35

Table 2: Association between alignment change $A(\ell)$ and agreement change $\kappa(\ell) - \kappa(\text{L3})$. Full-data results are descriptive. Cross-fitting estimates the two quantities on disjoint query folds; the reported exact p-value permutes whole model blocks, preserving eight cross-fitted contrasts per model.

Analysis	n	ρ	exact p
Full data	28	−0.799	–
Fold 0 → Fold 1	28	−0.684	–
Fold 1 → Fold 0	28	−0.685	–
Combined cross-fit	56	−0.683	0.0187

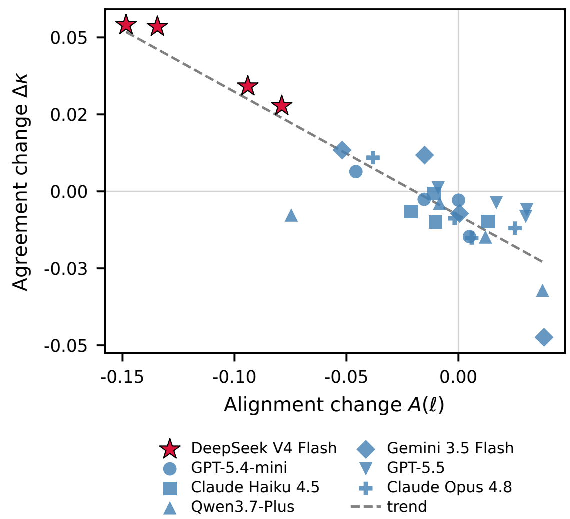

Figure 1: Full-data descriptive association across 28 model–tone cells ($\rho = -0.799$). DeepSeek (red) has the largest shifts, but the association remains negative without it ($\rho = -0.680$). Query-disjoint cross-fitting yields $\rho = -0.683$ with exact model-block $p = 0.0187$ (Table 2).

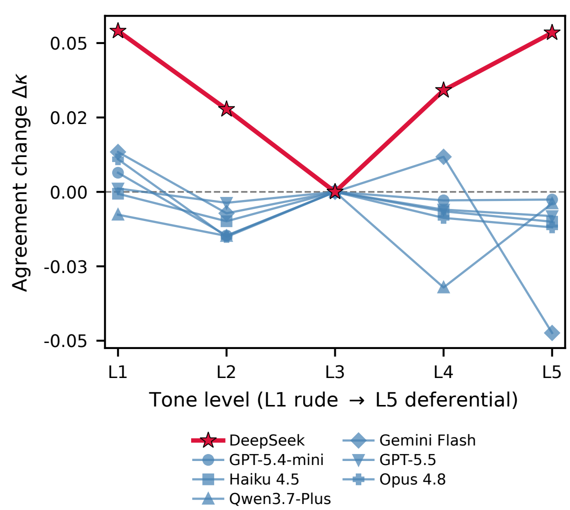

Figure 2: Change in agreement $\Delta\kappa$ across all five tone levels. DeepSeek (red) traces a structured U-shape. Other judges show smaller or unstructured trajectories; Gemini's L5 drop is driven by one anomalous paraphrase. Table 1 reports the two extremes.

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